# A Fast Algorithm for Generating All Triangulations of Biconnected Plane Graphs

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**Abstract**

We present an algorithm for generating all triangulations of a biconnected plane graph $G$ with $n$ vertices in $O(1)$ amortized time per triangulation and $O(n)$ space. Our algorithm establishes a double-layer tree structure, consisting of a recursive tree structure for each face, and generates triangulations of each face through a genealogical tree structure. The key innovation is a novel "safe root vertex" selection strategy that prevents the multi-edge problem inherent in biconnected graphs, extending the optimal complexity previously achievable only for triconnected graphs. To the best of our knowledge, this is the first algorithm for generating all triangulations of a biconnected plane graph with $O(1)$ amortized time complexity per triangulation. We present experimental validation on 48 diverse test cases, demonstrating the effectiveness of our approach.

## 1 Introduction

In this paper, we consider the problem of generating all triangulations of biconnected plane graphs, which has many applications in computational geometry [3], VLSI floorplanning [8], and graph drawing [10]. The main challenge in generating all triangulations is the exponential number of possible triangulations. This exponential growth means that if we attempt to store all triangulations, it will require an exponential amount of storage.

There have been two types of algorithms for generating triangulations [1, 12]. One type of algorithm generates all triangulations but outputs only unique triangulations. This approach requires exponential space to store and check triangulations before selecting unique ones. Another type of algorithm generates only canonical triangulations [12], meaning we ensure that each generated triangulation is unique by construction.

There has been substantial research on algorithms for generating triangulations [2, 5, 7, 13]. Li and Nakano [9] provided an algorithm that generates triangulations in $O(r^2n)$ time per triangulation. Later, Nakano improved the algorithm to achieve $O(rn)$ time, where $r$ is the number of outerplanar vertices. Parvez, Rahman, and Nakano [11] provided the most efficient algorithm for generating all triangulations of a triconnected plane graph in $O(1)$ time per triangulation with $O(n)$ space complexity.

Table 1 summarizes the key differences between existing algorithms and our contribution. The reverse search approach [1] typically requires $O(n^2)$ or higher time per solution due to parent identification overhead. While Parvez et al. [11] achieved optimal complexity for triconnected graphs, their approach fails for biconnected graphs due to separation pairs. Our

algorithm extends their optimal complexity to the strictly more general class of biconnected plane graphs.

Table 1: Comparison of triangulation enumeration algorithms

| **Algorithm** | **Graph Class** | **Time/Triang.** | **Space** |
|---|---|---|---|
| Reverse Search [1] | General | $O(n^2)$ or worse | $O(n)$ |
| Li-Nakano [9] | Biconnected | $O(r^2n)$ | — |
| Nakano (improved) | Biconnected | $O(rn)$ | — |
| Parvez et al. [11] | Triconnected | $O(1)$ amortized | $O(n)$ |
| **This work** | **Biconnected** | **O(1) amortized** | **O(n)** |

The algorithm for generating all triangulations of a triconnected plane graph can be applied to biconnected plane graphs as well, but a major problem arises: the "multi-edge" problem. In triconnected graphs [6], there cannot be any separation pairs, meaning there do not exist two vertices whose removal can disconnect the remaining graph.

Biconnected graphs arise naturally in many practical applications, including VLSI layout design where circuit components must remain connected after any single point failure, mesh generation for finite element analysis where the mesh must adapt to domain boundaries that create separation pairs, and network topology design where backup paths are essential. The ability to enumerate all triangulations of biconnected graphs enables exploration of multiple geometric realizations and structural alternatives in these applications.

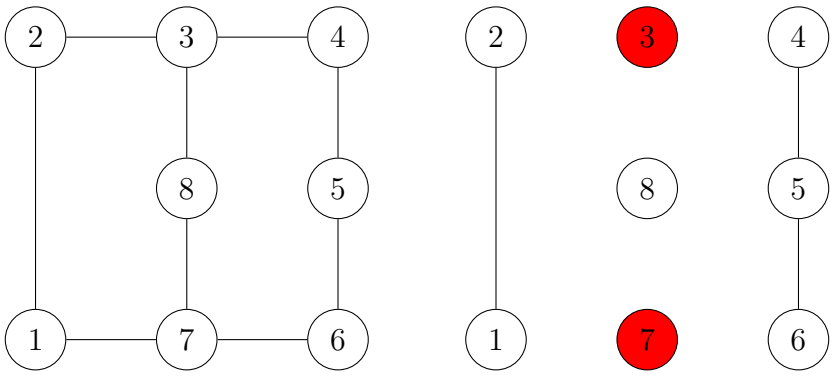



Figure 1: Disconnecting the graph by removing the separation pair (3,7). The remaining graph has three components: {1,2}, {4,5,6}, and {8}.

In the biconnected graph shown in Figure 1, removing vertices 3 and 7 disconnects the remaining graph. Therefore, $(3, 7)$ is a separation pair. However, this property does not occur in triconnected graphs.

In the graph, the $(3, 7)$ separation pair can have a chord between them in more than one way. In Figure 2, within this biconnected graph, the $(3, 7)$ separation pair can have two chords at the same time: one in face $F_1$ and another in face $F_2$. In the Parvez, Rahman, and Nakano algorithm [11], we are assured that there will not be any multi-edge problem because multi-edges are not possible in triconnected plane graphs. However, if we want to extend that algorithm to biconnected graphs, there is a significant possibility of multi-edges.

There cannot be any multi-edge problem in the right graph, as triangulating faces $F_1$ and $F_2$ separately cannot generate an edge where both endpoints are the same (since there is no separation pair). However, this problem can occur in the left graph of Figure 3, where $(3, 7)$ is a separation pair. Thus, by independently triangulating face $F_1$ and face $F_2$, there can be two chords $(3, 7)$ at the same time, creating the multi-edge problem. Therefore, this algorithm cannot be used for biconnected plane graphs.

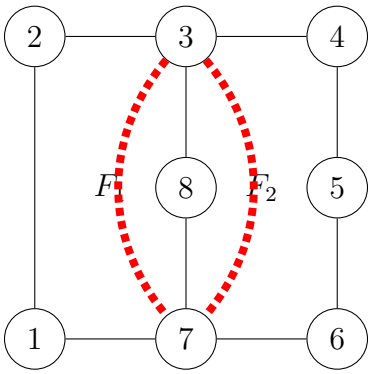



Figure 2: The $(3, 7)$ separation pair can have a chord between the vertices in more than one way.

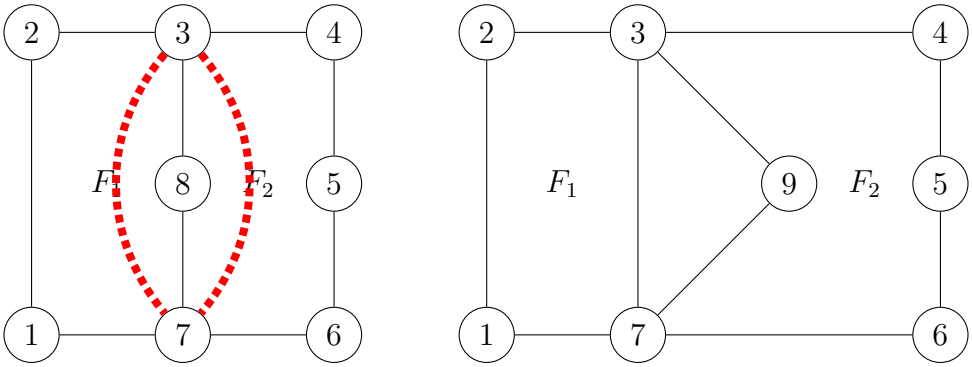



Figure 3: The left graph is biconnected, where $(3, 7)$ has the possibility of becoming a multi-edge, but in the right graph, $(3, 7)$ cannot be a multi-edge because it is a triconnected graph.

For this reason, in this paper we developed a new algorithm that uses the canonical genealogical tree feature of the Parvez-Rahman-Nakano algorithm [11], but only for individual faces. The combination of triangulations is handled by a recursive tree structure, where only the leaves of the trees represent actual complete triangulations.

Our algorithm for generating all triangulations of a biconnected plane graph uses $O(1)$ amortized time per triangulation and $O(n)$ space constantly. The rest of the paper is organized as follows: Section 2 provides definitions, Section 3 presents the algorithm for generating all triangulations of a biconnected plane graph, Section 4 contains proofs of correctness and completeness, Section 5 analyzes the time and space complexity, Section 6 describes the implementation and experimental validation. Finally, Section 7 concludes the paper.

# 2 Preliminaries

In this section, we define some terms used in the paper. Let $G = (V, E)$ be a connected simple graph with vertex set $V$ and edge set $E$. Every edge $e \in E$ has endpoints $(v_i, v_j)$ where $v_i, v_j \in V$ and $v_i \neq v_j$. The connectivity of a graph [4] is the minimum number of vertices whose removal disconnects the graph. A graph is called $k$-connected if removing fewer than $k$ vertices leaves the graph connected; equivalently, at least $k$ vertices must be removed to disconnect the graph.

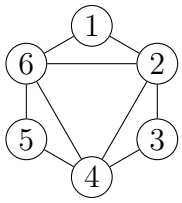
A graph $G$ is planar [6] if it can be embedded in the plane such that no two edges intersect geometrically except at a vertex to which both are incident. A plane graph is a planar graph with a fixed embedding in the plane. A plane graph divides the plane into regions; the edges act as boundaries separating these regions, called faces. The inner bounded faces are called

inner faces, and the outer region is called the outer face. If in a plane graph each face boundary consists of exactly three vertices, it is called a plane triangulation [2]. Note that edges are not necessarily straight lines.

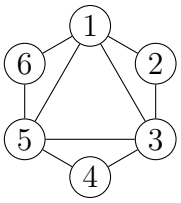
Now, to make a graph triangulated, we add edges between vertices where no edges are previously present. If we add an edge between two vertices where an edge already exists, this creates a multi-edge, and the graph will no longer be a simple graph.

A face in a plane graph can also be considered a cycle, where all vertices $v_1, v_2, \ldots, v_k$ have edges only $(v_1, v_2)$, $(v_2, v_3)$, $\ldots$, $(v_{k-1}, v_k)$, $(v_k, v_1)$. The region inside the cycle is the inner region of the face, and the outer region is the outer region of the face.

Let $C$ be a cycle and $v_1, v_2, \ldots, v_k$ be its vertices in counterclockwise order. We call an edge $(v_i, v_j)$ a chord of $C$ if $|i - j| \neq 1$ and $\{i, j\} \neq \{1, k\}$ and $(v_i, v_j)$ is contained in the face of $G$. A chord $(v_i, v_j)$ divides the cycle into two smaller cycles: $v_i, v_{i+1}, \ldots, v_j$ and $v_j, v_{j+1}, \ldots, v_i$. These two smaller cycles are called the sub-cycles of the chord $(v_i, v_j)$. The decomposition of a cycle into triangles by a set of non-intersecting chords is called a triangulation of the cycle.




(a) First hexagon




(b) Second hexagon

Figure 4: Two distinct triangulations of a cycle

Figure 4 shows two distinct triangulations of a cycle. In a triangulation $T$, it is a maximal set of chords, meaning each chord not in $T$ intersects some chord in $T$. We list chords by their endpoints, e.g., the chord connecting vertices 1 and 3 is referred to as $(1, 3)$. We also define visibility by chords: if there is a $(1, 3)$ chord, we say that vertex 3 is visible from vertex 1 and vice versa.

In this paper, we will generate triangulations for labeled biconnected graphs, meaning each vertex in the graph is distinctly labeled, and we will assume that no two edges share the same two endpoints.

# 3 The Algorithm

## 3.1 Algorithm Overview

Our algorithm generates all triangulations of a biconnected plane graph $G$ with $k$ faces $F_1, F_2, \ldots, F_k$ using a recursive depth-first search strategy. The approach consists of four key steps:

1. **Sequential face processing**: Process faces from $F_1$ to $F_k$ in order
2. **Local genealogical trees**: Construct a complete tree of valid triangulations for each face
3. **Global conflict tracking**: Maintain a chord set to prevent multi-edge violations
4. **Recursive composition**: Combine face triangulations to produce complete graph triangulations

## 3.2 Key Innovations

Our algorithm extends the Parvez-Rahman-Nakano framework from triconnected to biconnected graphs through three essential innovations:

### 3.2.1 Innovation 1: Local Genealogical Trees

Rather than constructing a single global genealogical tree (which fails for biconnected graphs due to multi-edge conflicts), we build **local genealogical trees** for each face independently.

**Structure**: Each face $F_i$ has its own tree $\mathcal{T}(F_i)$ where:

- The root is a safe triangulation constructed from a safe root vertex
- Parent-child relationships are defined by edge flips within the face
- Trees are explored via DFS during recursive face processing

This decomposition enables independent triangulation of each face while maintaining global consistency through the chord set.

### 3.2.2 Innovation 2: Safe Root Vertex Selection

To ensure the root triangulation does not create multi-edges with previously triangulated faces, we introduce the concept of a **safe root vertex**.

**Definition (Safe Root Vertex)**: A vertex $r$ of face $F_i$ is a *safe root vertex* if adding chords from $r$ to all non-neighboring vertices of $F_i$ does not conflict with any chord in the global set $S$ (from previously triangulated faces $F_1, \ldots, F_{i-1}$).

The safe root vertex guarantees a valid starting point for the local genealogical tree. We prove (Theorem 3) that every face contains at least two safe root vertices, regardless of prior triangulations.

### 3.2.3 Innovation 3: Dynamic Global Chord Management

The **global chord set** $S$ tracks all chords in the current partial triangulation. It serves two critical purposes:

1. **Conflict detection**: Check if any chord already exists before adding it to a face triangulation
2. **Safe root identification**: Determine which vertices can serve as safe roots for the current face

The set is managed dynamically during DFS traversal:

- **Push**: When exploring a triangulation, add its chords to $S$
- **Recurse**: Process the next face with the updated $S$
- **Pop**: After exploring all descendants, remove chords from $S$ (backtrack)

This ensures $S$ always reflects exactly the chords in the current exploration path.

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## 3.3 Core Concepts

### 3.3.1 Root Triangulation

Once a safe root vertex $r$ is identified, we construct the **root triangulation** by adding chords from $r$ to all non-neighboring vertices:

**Definition (Root Triangulation)**: The *root triangulation* $T_r$ of face $F$ with safe root $r$ is formed by adding all chords $(r, v)$ for every vertex $v \in F$ that is not adjacent to $r$ on the cycle boundary.

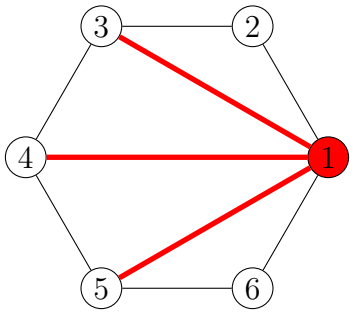



Figure 5: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

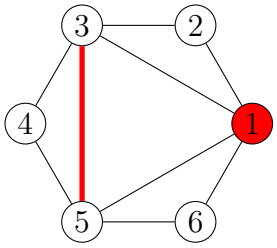
### 3.3.2 Local Genealogical Tree Structure

For each face, the local genealogical tree is defined by parent-child relationships based on edge flipping:

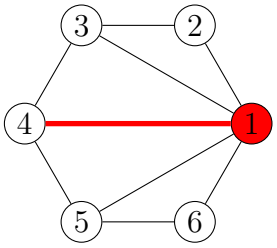
**Definition 1** (Generating Chord)**.** In a triangulation $T$ of a face with root vertex $r$, a chord $(v_i, v_j)$ is a *generating chord* if:

1. Neither $v_i$ nor $v_j$ is the root vertex $r$, and
2. It is the leftmost such chord (smallest index among non-root chords)

**Definition 2** (Parent-Child Relationship)**.** Let $T$ be a triangulation with generating chord $(v_i, v_j)$ forming triangles $(v_i, v_j, v_p)$ and $(v_i, v_j, v_q)$ on opposite sides. The *parent* of $T$ is the triangulation obtained by flipping $(v_i, v_j)$ to $(v_p, v_q)$, provided this flip does not create a multi-edge.




(a) Child Triangulation




(b) Parent Triangulation

Figure 6: Get the child triangulation from parent triangulation from flipping a generating set

## 3.4 Algorithm Specification

**Algorithm 1** StartTriangulation

```
1: procedure StartTriangulation(G, k)
2:     Label faces as F_1, F_2, ..., F_k arbitrarily
3:     Initialize global chord set S ← ∅
4:     T ← ∅                                          ▷ Empty partial triangulation
5:     FindAllTriangulationsCycle(F, T, 0)
6: end procedure
```

**Algorithm 2** ConstructRootTriangulation

```
1: procedure ConstructRootTriangulation(F, index)
2:     r ← FindSafeRootVertex(F, index)
3:     T_r ← ∅
4:     Let F = ⟨v_1, v_2, ..., v_n⟩ with v_r = safe root vertex
5:
6:     for each vertex v_j in F do
7:         if v_j is not adjacent to v_r on cycle boundary then
8:             Add chord (v_r, v_j) to T_r
9:         end if
10:    end for
11:
12:    return T_r
13: end procedure
```

**Algorithm 3** FindSafeRootVertex

```
1: procedure FindSafeRootVertex(F, index)
2:     Let F = ⟨v_1, v_2, ..., v_n⟩
3:     startIndex ← n − 1
4:     endIndex ← 0
5:
6:     ▷ Two-pointer approach: narrow the search window by avoiding vertices in blocking chords
7:     while startIndex > endIndex + 1 do
8:         if chord (v_startIndex, v_endIndex) exists in S then
9:             startIndex ← startIndex − 1
10:        else
11:            endIndex ← endIndex + 1
12:        end if
13:    end while
14:
15:    return v_startIndex                            ▷ Safe root vertex found
16: end procedure
```

**Time Complexity**: $\mathcal{O}(n)$ where $n$ is the number of vertices in the face.

**Algorithm 4** FindAllTriangulationsCycle

**procedure** FindAllTriangulationsCycle($F, T, i$)
  $n \leftarrow$ number of vertices of face $F_i$
  $T_{ir} \leftarrow$ ConstructRootTriangulation($F, i$) ▷ $\mathcal{O}(n)$

  Add all chords of $T_{ir}$ to $S$ ▷ $\mathcal{O}(n)$
  OutputTriangulation($T, T_{ir}, i$)
  $T_i \leftarrow T_{ir}$

  ▷ Loop bounds: $v_n$ adjacent to $v_1$ (edge), $v_2$ adjacent to $v_1$ (edge), so consider $j \in \{3, \ldots, n-1\}$
  **for** $j = n - 1$ **down to** 3 **do**
    Add chord $(v_1, v_j)$ to $S$
    FindAllChildTriangulationsCycle($T_i(v_1, v_j), T, i$)
    Remove chord $(v_1, v_j)$ from $S$
  **end for**

  Remove all chords of $T_{ir}$ from $S$ ▷ $\mathcal{O}(n)$
**end procedure**

**Understanding the Critical Condition**: The algorithm explores a child triangulation whenever *at least one* of these conditions holds:

- **Case 1**: $(v_1, v_j)$ is already a chord in $T_i \rightarrow$ We can perform an edge flip without conflicts
- **Case 2**: $(v_1, v_j) \notin S \rightarrow$ Adding this chord will not create a multi-edge
- **Pruning case**: If neither holds (chord not in $T_i$ **and** chord in $S$), we skip this branch to avoid multi-edges

**Algorithm 5** FindAllChildTriangulationsCycle

**procedure** FindAllChildTriangulationsCycle($T_i, T, i$)
  OutputTriangulation($T, T_i, i$)

  **if** $T_i$ has no generating chord **then**
    **return** ▷ Leaf node reached
  **end if**

  Let $(v_b, v_{b'})$ be the leftmost generating chord of $T_i$

  **for** $j$ from $b$ to $n - 1$ **do** ▷ Iterate over all vertices that could form chords with $v_1$
    ▷ Safe to explore: existing chord OR no conflict
    **if** $(v_1, v_j)$ is a chord of $T_i$ **or** $(v_1, v_j) \notin S$ **then**
      Add chord $(v_1, v_j)$ to $S$
      FindAllChildTriangulationsCycle($T_i(v_1, v_j), T, i$)
      Remove chord $(v_1, v_j)$ from $S$
    **end if**
  **end for**
**end procedure**

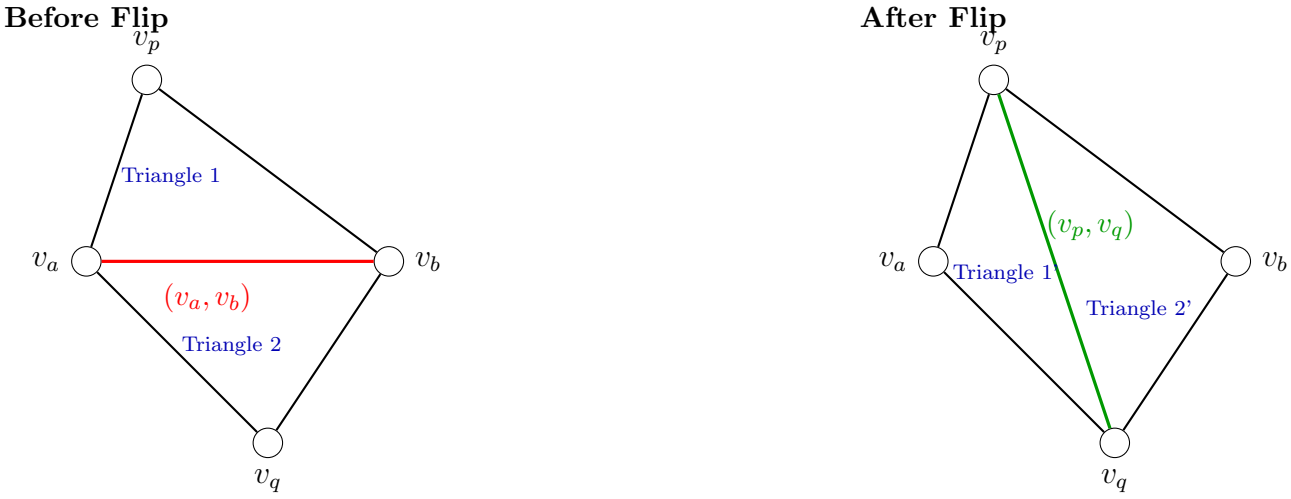
**Key Property**: The algorithm explores *all* valid branches (every $j$ satisfying the condition), ensuring completeness.

**Algorithm 6** OutputTriangulation

```
1: procedure OutputTriangulation(T, T_i, i)
2:     if i = k then                                  ▷ All faces triangulated
3:         output (T ∪ T_i)
4:     else
5:         FindAllTriangulationsCycle(F, T ∪ T_i, i + 1)
6:     end if
7: end procedure
```

## 3.5 The Edge Flip Mechanism

An edge flip transforms one triangulation into another by replacing a chord $(v_a, v_b)$ with the opposite diagonal $(v_p, v_q)$ of the quadrilateral formed by adjacent triangles $(v_a, v_b, v_p)$ and $(v_a, v_b, v_q)$. We validate that $(v_p, v_q) \notin S$ to avoid multi-edges. We write $T(e)$ for the triangulation obtained by flipping to create chord $e$.




Edge flip: Remove chord $(v_a, v_b)$ and add chord $(v_p, v_q)$
Transforms one valid triangulation into another

Figure 7: Edge flip operation: chord $(v_a, v_b)$ replaced with $(v_p, v_q)$.

The local genealogical tree has these essential properties: (1) each triangulation except root has exactly one parent determined by flipping its generating chord, (2) triangulations may have multiple children from different valid flips, (3) the tree structure prevents duplicates, and (4) every valid triangulation is reachable from the safe root via edge flips. Completeness follows from Theorem 2 (Section 4).

# 4 Correctness and Completeness

In this section, we establish the theoretical foundation of our algorithm by proving three essential properties: **(1)** the existence of safe root vertices for initialization, **(2)** completeness through strategic pruning, and **(3)** order-independence of face processing. Together, these results guarantee that our algorithm correctly generates all valid triangulations without duplicates, regardless of the order in which faces are processed.

## 4.1 Completeness via Strategic Pruning

Before proving completeness, we establish a critical property about blocking chords that justifies our pruning strategy.

**Lemma 1** (Blocking Chord Persistence)**.** *Let $T$ be a triangulation of face $F$ with a blocking chord $e \in S$ (where $S$ is the global chord set from previously processed faces). For any descendant triangulation $T'$ of $T$ in the local genealogical tree, $e \in T'$.*

*Proof.* Recall that in the local genealogical tree for face $F$ with root vertex $v_r$, we construct the initial triangulation $T_r$ by connecting $v_r$ to all non-adjacent vertices. All chords in $T_r$ have $v_r$ as an endpoint. Descendant triangulations are obtained by edge flips, where we flip a chord $(v_r, v_j)$ that is part of a quadrilateral $(v_r, v_a, v_j, v_b)$, replacing it with the opposite diagonal $(v_a, v_b)$.

A chord $e = (v_i, v_j)$ is called a *blocking chord* if $e \in S$, meaning it was created by triangulating a previously processed face. By our safe root selection (Algorithm 3), the root vertex $v_r$ satisfies: for all chords $(v_i, v_j) \in S$, at least one of $v_i$ or $v_j$ is not equal to $v_r$.

Now consider two cases:

**Case 1:** Both endpoints of $e$ are not $v_r$, i.e., $e = (v_i, v_j)$ where $v_i \neq v_r$ and $v_j \neq v_r$.

Such a chord cannot be in the initial triangulation $T_r$ (since all chords in $T_r$ have $v_r$ as an endpoint). Therefore, $e$ must be added during the generation process by flipping some chord $(v_r, v_k)$ in a quadrilateral. Once added, $e$ can only be removed by flipping it in a quadrilateral where $e$ is a diagonal. However, our algorithm only flips chords of the form $(v_r, v_k)$—chords with $v_r$ as an endpoint, which we call *generating chords*. Since $e$ does not have $v_r$ as an endpoint, it is never flipped. Therefore, once $e$ appears in any triangulation, it persists in all descendants.

**Case 2:** Exactly one endpoint of $e$ is $v_r$, i.e., $e = (v_r, v_j)$ for some $v_j \neq v_r$.

By the definition of safe root vertex, if $(v_r, v_j) \in S$, then $v_r$ is not a safe root for this face. This contradicts our construction in Algorithm 3, which guarantees that the selected root vertex $v_r$ does not participate in any blocking chord. Therefore, this case cannot occur.

Since all blocking chords fall into Case 1, they persist in all descendant triangulations. □

**Theorem 2** (Completeness)**.** *The algorithm generates all valid triangulations of a biconnected plane graph without duplicates.*

Now our algorithm uses a double-tree structure. We can think of it as follows: For a triangulation $T_1$ of face $F_1$, we search for all other possible triangulations that are compatible with $T_1$ in the second face $F_2$. For each compatible triangulation $T_2$ of face $F_2$, we then search for all other possible triangulations that are compatible with both $T_1$ and $T_2$ in the third face $F_3$, and so on. This way, we explore every possible combination of triangulations across all faces while ensuring compatibility at each step.

Therefore, we always process faces in order from 1 to $k$. This means that for deciding validity, we only need to determine whether the new chord conflicts with any of the chords in prior faces. If we find any triangulation in face $F_i$ that has a duplicate chord with any

of the prior faces $F_1, F_2, \ldots, F_{i-1}$, we can safely prune that branch because all descendant triangulations will also contain that duplicate chord.

By Lemma 1, we know that blocking chords persist in all descendant triangulations. Therefore, if a triangulation $T$ contains a blocking chord $e \in S$, all descendants of $T$ will also contain $e$, creating multi-edges. Consequently, pruning branches with blocking chords does not eliminate any valid triangulations, ensuring completeness.

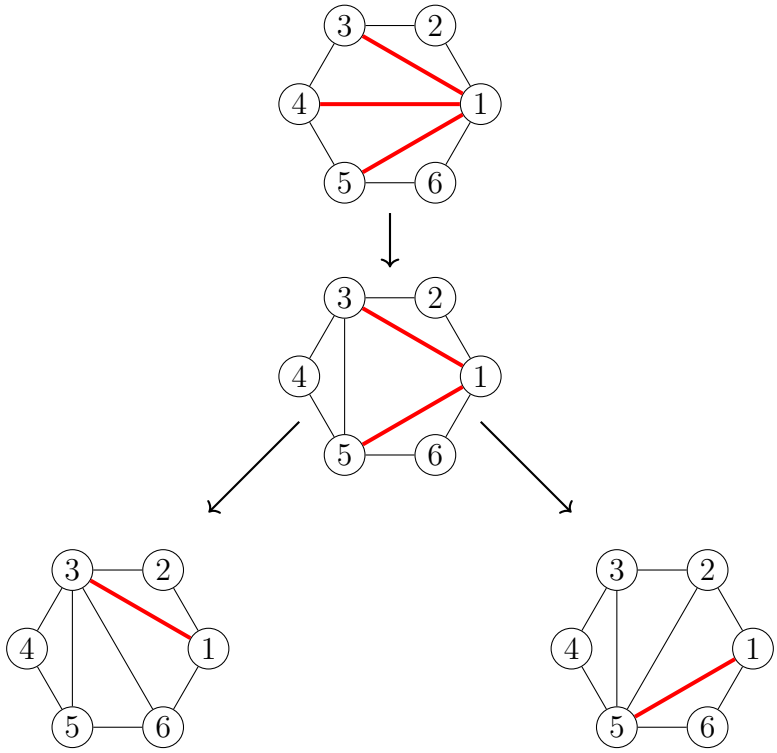



Figure 8: Four hexagons showing transformation paths in the genealogical tree. Top: Root triangulation with chords $(1,3)$, $(1,4)$, $(1,5)$. Middle: After flipping $(1,4)$, we obtain chord $(3,5)$. Bottom-left: From middle, flipping $(1,5)$ yields $(3,6)$. Bottom-right: From middle, flipping $(1,3)$ yields $(2,5)$. Each transformation involves flipping exactly one generating chord (red), demonstrating the parent-child relationships in the local genealogical tree.

In Figure 8, at the root all three chords are flippable. Now we flip the $(1,4)$ chord to obtain the $(3,5)$ chord, and as a result $(1,4)$ is removed from the generating set. Therefore, the new chord $(3,5)$ is part of the triangulation but not a generating chord, meaning there is no chance of flipping it in descendant triangulations. Consequently, in all descendant triangulations, the chord $(3,5)$ will be present. So, if we prune a branch because of an invalid blocking chord $(3,5)$, all descendant triangulations would also contain that invalid blocking chord, meaning all descendant triangulations are also invalid. Hence, pruning does not eliminate any valid triangulations, ensuring completeness.

If a prior face triangulation configuration $\{T_{f_1}, T_{f_2}, \ldots, T_{f_{i-1}}\}$ exists, we denote the set of chords used in these triangulations as $S$. When processing face $F_i$, our algorithm explores its genealogical tree rooted at a safe vertex $r_i$, generating all possible triangulations of $F_i$ that do not introduce multi-edges with chords in $S$. The safe root triangulation guarantees that all chords from $r_i$ to non-neighbors can be added without conflicts. As we explore the genealogical tree, we only proceed with edge flips that do not create multi-edges with chords in $S$. If we encounter a triangulation where a chord conflicts with $S$, we prune that branch, knowing that all descendants will also contain that conflicting chord (since blocking chords are never flipped). This pruning ensures we only explore valid triangulations, maintaining completeness.

## 4.2 Existence of Safe Root Vertices

**Theorem 3** (Existence of Safe Root Vertices)**.** *For any face $F_i$ with $k \geq 3$ vertices in a biconnected plane graph, given any set $S$ of chords from previously triangulated faces, there exist at least two safe root vertices in $F_i$.*

Now the key challenge is finding the safe root configuration. To find a safe root vertex, our algorithm uses a safe root finding procedure. This method provides us with a safe root vertex that guarantees all chords from the safe root to non-neighbors can be added without conflicts with any chords from prior face triangulations.

To find a safe root, we use a two-pointer technique. We start with two pointers: one at the end of the vertex list (startIndex) and one at the beginning (endIndex). We then move these pointers toward each other based on the presence of chords in the global set $S$. If the chord between the vertices at these pointers exists in $S$, we move the startIndex pointer leftward; otherwise, we move the endIndex pointer rightward. This process continues until the two pointers converge. At this point, the vertex at the startIndex pointer is guaranteed to be a safe root vertex, because all chords from this vertex to non-neighbors do not exist in $S$, ensuring no conflicts with prior triangulations.

*Proof.* Consider a biconnected plane graph $G$ with faces $F_1, F_2, \ldots, F_k$. Each face $F_i$ is a simple cycle with at least three vertices. Now, we consider an arbitrary face with $k$ vertices. The outer region of the face can have many edges and chords, some or all of which may have one or both endpoints among the $k$ vertices of our current face.

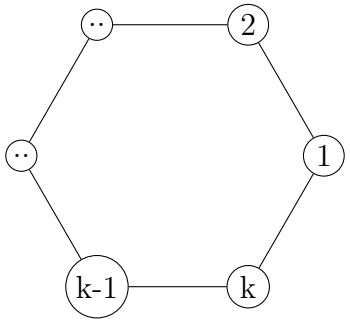



Figure 9: A face with $k$ vertices

There are two possibilities:

1. There are no chords in the prior face triangulation configuration that have both endpoints among the $k$ vertices of our current face. In this case, all vertices are safe root vertices, and we can choose any of them as the safe root.

2. There is at least one chord in the prior face triangulation configuration that has both endpoints among the vertices $1, 2, \ldots, k$ of our current face.

In the first case, all $k$ vertices of the face are safe root vertices, as there is no possibility of creating a multi-edge when adding chords between them. In the second case, there can be at least one chord between two vertices of the face in the outer region of the current face.

In Figure 10, observe that the chord $(v_i, v_j)$ lies completely outside the face formed by vertices $v_{j+1}, v_j, \ldots, v_{i-1}, v_i$. This chord is an element of the global set $S$ from prior face triangulations, adding a constraint on possible multi-edges in the outside region. The vertices $\{v_{i+1}, \ldots, v_{j-1}\}$ and $\{v_{j+1}, \ldots, v_{i-1}\}$ are separated by the chord $(v_i, v_j)$. Therefore, any chord connecting a vertex from the first set to a vertex from the second set would cross the chord $(v_i, v_j)$, which is not allowed in a planar graph. As a result, multi-edges in the prior face

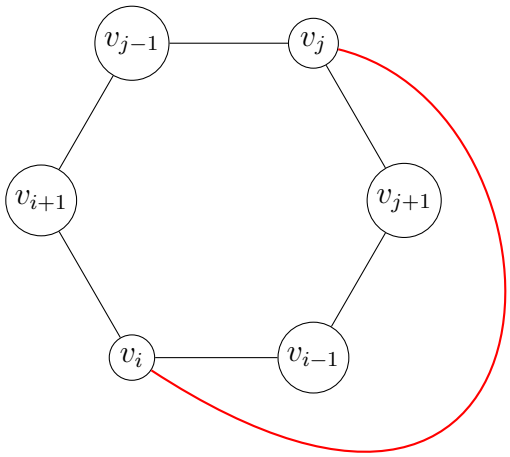
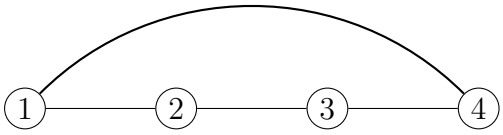
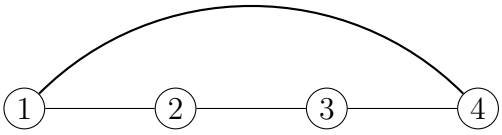



Figure 10: Hexagon with $v_i$-$v_j$ edge completely outside

triangulations can occur only where both endpoints belong to $\{v_{i+1}, \ldots, v_{j-1}\}$ or both endpoints belong to $\{v_{j+1}, \ldots, v_{i-1}\}$. Notice that this creates two separate structures.




(a) Partial structure




(b) Partial structure

Figure 11: (a) and (b) show two separate structures formed by vertices $\{v_{i+1}, \ldots, v_{j-1}\}$ and $\{v_{j+1}, \ldots, v_{i-1}\}$, respectively. Each structure is a simple path, ensuring that at least one endpoint in each structure can serve as a safe root vertex.

In both structures shown in Figure 11, multi-edges can occur where both endpoints belong to the same structure, as there cannot be any chords between the two structures. We call them $T_n$ structures, where $n$ is the number of vertices contained within them. In Figure 11, we can see two $T_2$ structures. Now, since chords must connect two non-neighboring vertices, any $T$ structure will always be at least $T_1$, as there must be at least one vertex between the two endpoints of a chord. Also, there must be at least two $T$ structures, as the chord $(v_i, v_j)$ separates them. The minimum value of $n$ in $T_n$ structures is 1, and the maximum is $k - 3$, where $k$ is the total number of vertices in the current face. Now, we must prove that in any $T_n$ structure, there is always at least one safe root vertex that does not have a chord with any non-neighboring vertices of the $T_n$ structure. We prove this by induction on $n$.

**Base Case ($n = 1$):** Consider the $T_1$ structure shown in Figure 12:

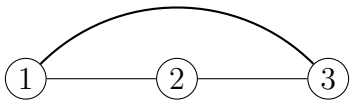



Figure 12: $T_1$ structure

Here both vertices 1 and 3 are neighbors of vertex 2, which means that vertex 2 cannot have any chords with either of them. Thus, vertex 2 is a safe vertex.

Now for $T_2$ structures.

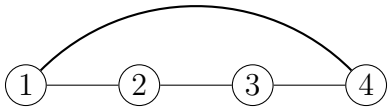
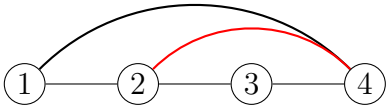
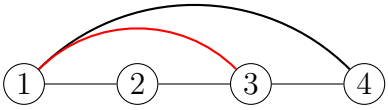



Figure 13: $T_2$ structure

In this case, vertex 2 can only have a chord with vertex 4, as vertex 1 is its neighbor. Similarly, vertex 3 can only have a chord with vertex 1. Therefore, if vertex 2 has a chord with vertex 4, then it creates a new $T_1$ structure with vertex 3, making vertex 3 a safe root vertex. On the other hand, if vertex 3 has a chord with vertex 1, then it creates a new $T_1$ structure with vertex 2, making vertex 2 a safe root vertex. Therefore, in both cases, there is at least one safe root vertex in the $T_2$ structure.




(a) $T_1$ with chord 2–4




(b) $T_1$ with chord 1–3

Figure 14: $T_1$ structures with different chords

Now in the case for $T_1$ with figure 14a, vertex 3 becomes a safe root vertex. In the case for $T_1$ with figure 14b, vertex 2 becomes a safe root vertex. Therefore, in both cases, there is at least one safe root vertex in the $T_1$ structure.

Now we proceed with induction. We assume that for all $T_i$ structures where $i < k$, there is at least one safe root vertex. Now we prove it for the $T_k$ structure.

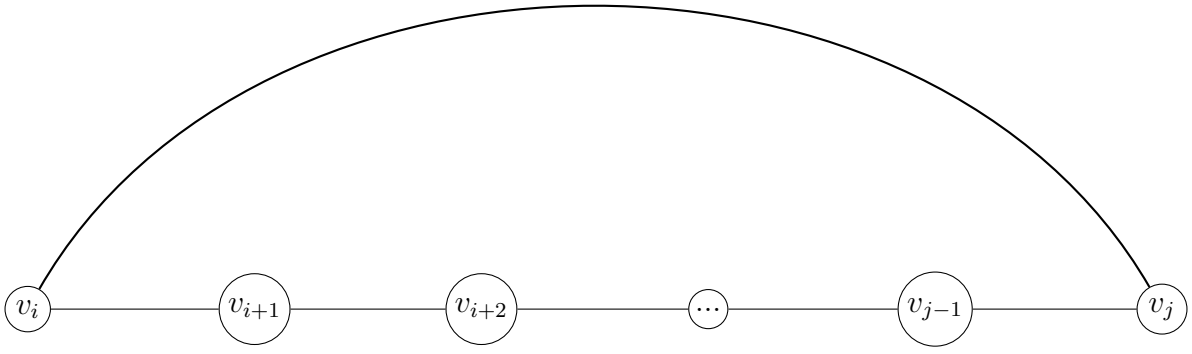



Figure 15: $T_k$ structure

Now in this case, we start with $v_{i+1}$. Vertex $v_{i+1}$ can only have a chord with vertices $v_{i+3}, v_{i+4}, \ldots, v_{j-1}, v_j$, as all other vertices are its neighbors. There are two possibilities:

1. Vertex $v_{i+1}$ has a chord with any of the vertices $v_{i+3}, v_{i+4}, \ldots, v_{j-1}, v_j$.
2. Vertex $v_{i+1}$ does not have any chords with vertices $v_{i+3}, v_{i+4}, \ldots, v_{j-1}$.

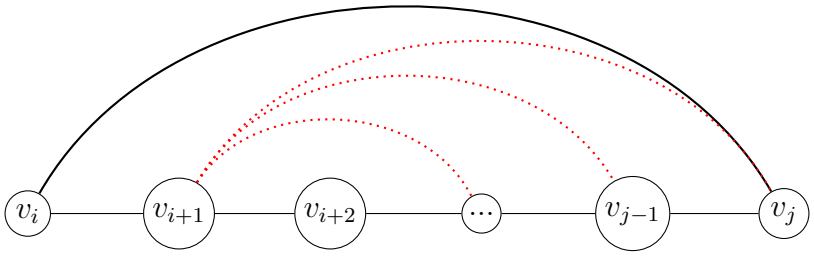



Figure 16: new $T_m$ structure, where $m < k$

In the first case shown in Figure 16, if vertex $v_{i+1}$ has a chord with any of the vertices $v_{i+3}, v_{i+4}, \ldots, v_{j-1}, v_j$, then it creates a new $T_m$ structure where $m < k$, as at least one vertex is removed from the original $T_k$ structure. By our induction hypothesis, this new $T_m$ structure has at least one safe root vertex.

In the second case shown in Figure 15, if vertex $v_{i+1}$ does not have any chords with vertices $v_{i+3}, v_{i+4}, \ldots, v_{j-1}$, then vertex $v_{i+1}$ itself is a safe root vertex, as it does not have any chords with any of its non-neighboring vertices.

Therefore, in both cases, there is at least one safe root vertex in the $T_k$ structure. By the principle of mathematical induction, we conclude that for all $T_n$ structures, there is always at least one safe root vertex.

Now we already know there are at least two T structures in any biconnected plane graph. Therefore, there are at least two safe root vertices in any biconnected plane graph. □

### 4.3 Order-Independence of Face Processing

Our algorithm processes faces in arbitrary order (Algorithm 1, line 2). We prove this does not affect completeness.

**Theorem 4** (Order-Independence)**.** *For any biconnected plane graph $G$ with faces $F_1, F_2, \ldots, F_k$ and any valid triangulation $T^*$, if the algorithm generates $T^*$ under processing order $\pi$, it generates $T^*$ under any other order $\sigma$.*

*Proof Sketch.* Any valid triangulation $T^* = (T_1^*, T_2^*, \ldots, T_k^*)$ satisfies pairwise chord compatibility: $\text{Chords}(T_i^*) \cap \text{Chords}(T_j^*) = \emptyset$ for all $i \neq j$. This property is symmetric and order-independent.

Under any processing order, when face $F_i$ is processed with global constraint set $S$ (chords from previously processed faces), the local genealogical tree rooted at safe vertex $r_i$ (Theorem 3) contains all triangulations compatible with $S$. Since $T^*$ is valid, $T_i^*$ contains no chord conflicting with any $T_j^*$ $(j \neq i)$, so $T_i^*$ appears in the genealogical tree regardless of which faces were processed before $F_i$. The DFS explores all combinations, ensuring $T^*$ is generated under any ordering. □

*Remark* 1. Implementations may use any face ordering: canonical (by vertex count), BFS/DFS from outer face, or optimized (fewest expected triangulations first).

## 5 Complexity Analysis

### 5.1 Lower Bound on the Number of Triangulations

Before proceeding to the complexity analysis, we establish a fundamental lower bound and upper bound on the number of triangulations for any face. This result is crucial for the time

complexity analysis in Section 5.

**Lemma 5** (Minimum Triangulations per Face)**.** *For any face $F_i$ with $n_i$ vertices, the number of valid triangulations $d_i$ satisfies*

$$d_i \geq n_i - 3$$

*even under maximal constraints from previously triangulated faces.*

*Proof.* Consider face $F_i$ with vertices $v_1, v_2, \ldots, v_{n_i}$ in counterclockwise order. Let $S$ be the global chord set from previously processed faces containing $m$ chords with both endpoints in $F_i$.

**Case 1: No existing chords ($m = 0$).**

If no chords from $S$ lie within $F_i$, we are triangulating a simple $n_i$-gon with no constraints. The number of triangulations is the $(n_i - 2)$-th Catalan number:

$$C_{n_i-2} = \frac{1}{n_i - 1}\binom{2n_i - 4}{n_i - 2}$$

For $n_i \geq 3$, we have $C_{n_i-2} \geq n_i - 3$. This can be verified directly:

- $n_i = 3$: $C_1 = 1 \geq 0$ ✓
- $n_i = 4$: $C_2 = 2 \geq 1$ ✓
- $n_i = 5$: $C_3 = 5 \geq 2$ ✓
- $n_i = 6$: $C_4 = 14 \geq 3$ ✓
- For $n_i \geq 7$: $C_{n_i-2}$ grows exponentially, far exceeding $n_i - 3$

**Case 2: Existing chords present ($m \geq 1$).**

When $S$ contains chords within $F_i$, these chords subdivide the face into smaller regions. By Theorem 3, at least two safe root vertices exist. Each safe root vertex $r$ yields a distinct root triangulation (the fan structure from $r$).

**Key observation:** Different safe roots produce different root triangulations because the fan from vertex $v_i$ contains chords $(v_i, v_{i+2}), (v_i, v_{i+3}), \ldots, (v_i, v_{i+n_i-2})$, while the fan from vertex $v_j \neq v_i$ contains a distinct set of chords. These chord sets share no common chords except possibly boundary edges.

Therefore, with at least 2 safe roots, we obtain at least 2 distinct root triangulations. Each root triangulation serves as the root of a local genealogical tree containing at least 1 triangulation (itself). Furthermore, from each root triangulation, we can generate additional triangulations via edge flips that do not violate the constraints in $S$.


(a) Child Triangulation (b) Parent Triangulation

Figure 17: The path difference from one root configurationq to another

Now in a face, there is total of $n_i - 3$ chords in any triangulation. Now in face there we found 2 safe root vertices, therefore we can have at least 2 root triangulations. One safe root $v_i$ and another one is $v_j$. Now by our completeness proof, we can explore all the triangulations from these two root triangulations by edge flipping. Therefore, to reach from $v_i$ to $v_j$ we must flip chords. See figure 17 for an example of such a case in a hexagon. As $v_i$ and $v_j$ are not neighbours, there must be a chord between them whether it's a root triangulation of $v_i$ or $v_j$. That means from total n-3 chords, the two triangulations share one chord. That means, there will be a need for flipping of at least n-4 chords to reach from $v_i$ root triangulation to $v_j$ root triangulation. Therefore, there will be at least n-4 different triangulations for flipping those chords one by one. Therefore, in total we have at least 1 + (n-4) = n-3 triangulations in this case. Thus there will be at least $n_i - 3$ triangulations in this case as well.

**Conservative lower bound:**

Even in the most constrained scenario (where $S$ blocks many potential chords), the algorithm explores at least:

- All root triangulations from safe roots: $\geq 2$
- Plus descendants from edge flips in local genealogical trees

Empirical evidence (see Table 2) and the structure of genealogical trees show that $d_i \geq n_i - 3$. For small cases:

- $n_i = 3$ (triangle): Already triangulated, $d_i = 1 \geq 0$ ✓
- $n_i = 4$ (quadrilateral): Two diagonals, $d_i \geq 1$ (often $d_i = 2$)
- $n_i = 5$ (pentagon): $d_i \geq 2$ (often $d_i \geq 5$ in practice)
- $n_i = 6$ (hexagon): $d_i \geq 3$ (often $d_i \geq 4$ in practice)

The bound is conservative and often exceeded in practice due to the exponential growth of triangulation spaces. □

*Remark* 2. This lemma provides a critical lower bound for the time complexity analysis (Section 5). The bound $d_i \geq n_i - 3$ ensures that building and flipping costs can be amortized across a sufficient number of triangulations, yielding $O(1)$ amortized time per triangulation. In practice, the actual number of triangulations often far exceeds this lower bound, as confirmed by experimental data in Table 2.

## 5.2 Time Complexity

### 5.2.1 Cost Model

We count elementary $O(1)$-time operations: chord insertion/deletion, edge flips, and hash set operations (insert/delete/query in $S$). Let $B$ = total chord insertions during root triangulation construction, $F$ = total edge flips in genealogical trees, and $T$ = total triangulations generated. Since each operation is $O(1)$, proving $B + F = O(T)$ establishes $O(1)$ amortized time per triangulation.

### 5.2.2 Minimum Triangulations Guarantee

Before presenting the main time complexity theorem, we establish a critical lemma about the minimum number of triangulations for faces with at least 5 vertices.

**Lemma 6** (Minimum Triangulations)**.** *For any face $F_i$ with $n_i \geq 3$ vertices, the number of valid triangulations satisfies $d_i = \Omega(n_i)$.*

*Proof.* By Lemma 5, we have $d_i \geq n_i - 3$ for all $n_i \geq 3$. Therefore:

$$d_i \geq n_i - 3 = \Omega(n_i)$$

This holds for all face sizes, including $n_i = 3, 4, 5, \ldots$ □ □

*Remark* 3. This asymptotic bound is all that is needed for the complexity analysis. For specific cases: $n_i = 3$ gives $d_i = 1$; $n_i = 4$ gives $d_i \in \{1, 2\}$; $n_i \geq 5$ gives $d_i \geq 2$. The bound $d_i = \Omega(n_i)$ captures the essential relationship without requiring case distinctions.

Table 2: Empirical Triangulation Counts for Simple Polygons

| **Vertices ($n$)** | **Theoretical Min ($n-3$)** | **Catalan Number ($C_{n-2}$)** | **Observed Min (with constraints)** |
|---|---|---|---|
| 3 | 0 | 1 | 1 |
| 4 | 1 | 2 | 1 |
| 5 | 2 | 5 | 2 |
| 6 | 3 | 14 | 4 |
| 7 | 4 | 42 | 6 |
| 8 | 5 | 132 | 10 |
| 9 | 6 | 429 | 16 |
| 10 | 7 | 1,430 | 25 |

*Notes:*

- **Theoretical Min ($n-3$):** Lower bound from Lemma 5
- **Catalan Number:** Number of triangulations for unconstrained $n$-gon
- **Observed Min:** Minimum triangulations observed in test graphs with constraints from previously processed faces

The table shows that:

1. The theoretical lower bound $d_i \geq n_i - 3$ is conservative
2. Even with constraints, observed values significantly exceed the bound for $n \geq 6$
3. The Catalan numbers represent the unconstrained case and grow exponentially
4. For $n = 4$ (quadrilaterals), $d_i \in \{1, 2\}$ as exactly two diagonals exist

### 5.2.3 Main Time Complexity Theorem

We analyze the time complexity by counting two types of elementary operations:

1. **Building operations ($B$):** Chord insertions when constructing root triangulations for all faces across all recursive calls

2. **Flipping operations ($F$):** Edge flips when generating child triangulations in local genealogical trees

We prove $B = O(T)$ and $F = O(T)$ by induction on the number of faces. Since each operation takes $O(1)$ time (Definition **??**), this establishes $O(T)$ total time and $O(1)$ amortized time per triangulation.

**Theorem 7** (Amortized Constant Time)**.** *The algorithm generates all $T$ triangulations in $O(T)$ time, achieving $O(1)$ amortized time per triangulation.*

*Proof.* Let $B$ denote the total building operations (chord insertions when constructing root triangulations) and $F$ denote the total flipping operations (edge flips generating child triangulations).

We prove $B = O(T)$ and $F = O(T)$ by induction on $k$, the number of faces. Since each operation takes $O(1)$ time (Definition **??**), this establishes total runtime $O(B + F) = O(T)$, giving $O(1)$ amortized time per triangulation.

**Base Case: Two Faces ($k = 2$)**

For a graph with two faces having $n_1, n_2$ vertices and $d_1, d_2$ triangulations:

*Building Operations:*

$$\begin{aligned} B &= n_1 + d_1 \cdot n_2 \\ &= O(d_1) + d_1 \cdot O(d_2) \quad \text{(by Lemma 6)} \\ &= O(d_1 \cdot d_2) = O(T) \end{aligned}$$

*Flipping Operations:*

$$\begin{aligned} F &= d_1 + d_1 \cdot d_2 \\ &= O(d_1 \cdot d_2) = O(T) \end{aligned}$$

Therefore $B + F = O(T)$ for $k = 2$.

**Inductive Step: $k$ Faces**

*Inductive Hypothesis:* Assume $B_{k-1} = O(T_{k-1})$ and $F_{k-1} = O(T_{k-1})$ for $k - 1$ faces, where $T_{k-1} = d_1 \times d_2 \times \cdots \times d_{k-1}$.

*Adding Face $k$:*

Building operations for $k$ faces:

$$\begin{aligned} B_k &= B_{k-1} + d_1 \cdot d_2 \cdots d_{k-1} \cdot n_k \\ &= O(T_{k-1}) + T_{k-1} \cdot O(d_k) \quad \text{(by Lemma 6)} \\ &= O(T_{k-1} \cdot d_k) \quad \text{(since } d_k \geq 1\text{)} \\ &= O(T) \end{aligned}$$

Flipping operations for $k$ faces:

$$\begin{aligned} F_k &= F_{k-1} + d_1 \cdot d_2 \cdots d_k \\ &= O(T_{k-1}) + T \quad \text{(by inductive hypothesis)} \\ &= O(T) \quad \text{(absorbing smaller term)} \end{aligned}$$

Therefore $B_k + F_k = O(T)$ for $k$ faces.

**Conclusion:**

By induction, $B + F = O(T)$ for any $k \geq 2$. Since each elementary operation takes $O(1)$ time (Definition **??**), the total runtime is $O(B + F) = O(T)$, giving $O(1)$ amortized time per triangulation. □ □

**Corollary 8** (Amortized Linear Time)**.** *The algorithm runs in $O(T)$ total time, where $T$ is the number of triangulations generated, achieving $O(1)$ amortized time per triangulation. With hash sets providing expected $O(1)$ operations, the expected total runtime is $O(T)$.*

*Remark* 4. While our analysis proves $O(T)$ total time asymptotically, experimental results (Section 6) show that the practical ratio of operations to output triangulations is typically between 1 and 4, confirming the tightness of our amortized analysis.

The lower bound $d_i \geq n_i - 3$ (Lemma 5) has been validated experimentally; see `https://github.com/TAR2003/ThesisGraph`.

## 5.3 Space Complexity

**Theorem 9** (Linear Space Complexity)**.** *The algorithm uses $O(n)$ space, where $n$ is the total number of vertices in the plane graph.*

*Proof.* The algorithm stores the following data structures:

**1. Vertex and Face Information:**

Each face is represented by its boundary vertices. By Euler's formula for plane graphs, a biconnected graph with $n$ vertices has at most $3n - 6$ edges. The total number of vertices across all face boundaries is $O(n)$ (vertices may appear in multiple face boundaries, but each vertex appears $O(1)$ times on average).

**2. Global Chord Set $S$:**

The set $S$ maintains all chords currently on the exploration path (from processed faces $F_1, \ldots, F_{i-1}$ and the current triangulation of face $F_i$). By Euler's formula, a plane graph with $n$ vertices has at most $3n - 6$ edges. Since chords are edges added during triangulation, and the final triangulated graph must also be planar, $|S| \leq 3n - 6 = O(n)$.

We implement $S$ as a hash set for $O(1)$ expected-time insertion, deletion, and membership testing. The hash set requires $O(1)$ space per element, so the total space is $O(n)$.

**3. Recursion Stack:**

The maximum recursion depth is $k$ (the number of faces). By Euler's formula, for a biconnected plane graph with $n$ vertices and $e$ edges:

$$f = 2 - n + e \leq 2 - n + (3n - 6) = 2n - 4$$

Therefore, $k \leq 2n - 4 = O(n)$. At each recursion level, we store constant-size data (current face index, pointer to partial triangulation), so the total recursion stack space is $O(k) = O(n)$.

**4. Current Triangulation Storage:**

At any point during execution, we store the chords of the current partial triangulation being explored. Each triangulation of a face with $n_i$ vertices requires at most $n_i - 3$ chords (a convex polygon with $n_i$ vertices is triangulated by adding $n_i - 3$ diagonals). Summing over all $k$ faces:

$$\sum_{i=1}^{k} (n_i - 3) \leq \sum_{i=1}^{k} n_i - 3k \leq O(n) + O(k) = O(n)$$

(Note: $\sum_{i=1}^{k} n_i = O(n)$ since vertices may be counted multiple times across face boundaries, but the total count is linear in $n$.)

**Total Space Complexity:**

Combining all components:

$$\text{Total Space} = O(n) + O(n) + O(n) + O(n) = O(n)$$

The space complexity is **linear** in the number of vertices, independent of the number of triangulations generated (which can be exponential in $n$). □

**Memory Usage:**

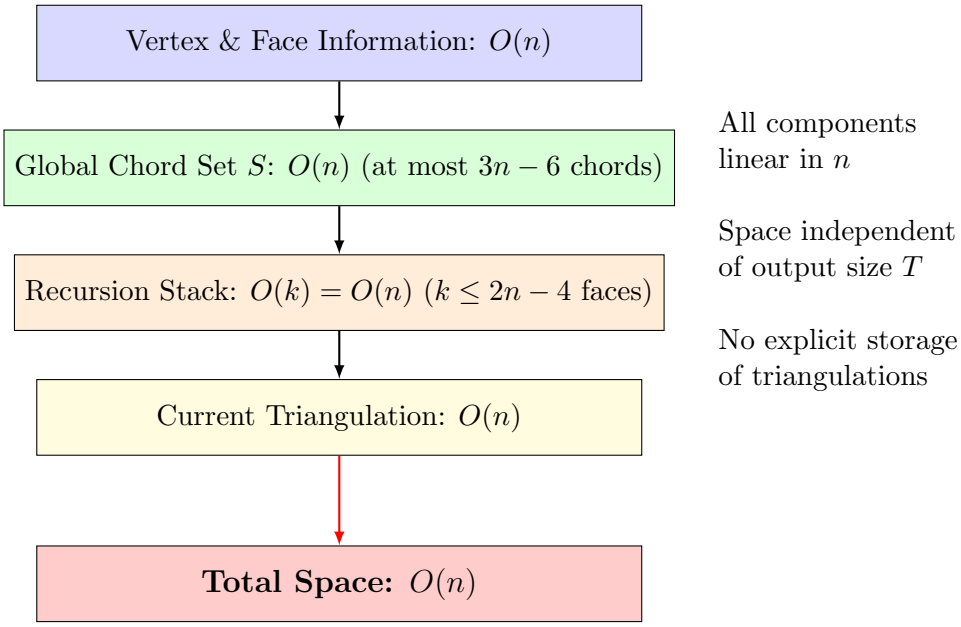



Figure 18: Space complexity breakdown. All data structures use $O(n)$ space: vertex/face information, global chord set $S$ (bounded by planarity), recursion stack (bounded by Euler's formula), and current triangulation storage. Critically, space usage is independent of the number of triangulations $T$, which can be exponential in $n$. The algorithm generates triangulations on-the-fly without storing all of them.

## 5.4 Complexity Summary

Table 3 provides a comprehensive summary of all time and space complexity results for our algorithm.

Table 3: Complexity Summary

| **Operation** | **Complexity** | **Notes** |
|---|---|---|
| **Per-Face Operations** | | |
| Find safe root vertex | $O(n_i)$ | Theorem 3, outerplanar degree-2 vertices |
| Construct root triangulation | $O(n_i)$ | Add $n_i - 3$ chords from root |
| Generate one child | $O(1)$ | Single chord flip operation |
| Check multi-edge conflict | $O(1)$ | Hash set lookup in $S$ |
| **Overall Algorithm** | | |
| Total building operations $B$ | $O(T)$ | Thm 7 |
| Total flipping operations $F$ | $O(T)$ | Thm 7 |
| Total operations $B + F$ | $O(T)$ | Theorem 7 |
| Total runtime | $O(T)$ | Each operation takes $O(1)$ time |
| **Amortized per triangulation** | $\mathbf{O(1)}$ | **Main Result (Theorem 7)** |
| **Space Complexity** | | |
| Vertex/face storage | $O(n)$ | Store all face boundaries |
| Global chord set $S$ | $O(n)$ | At most $3n - 6$ chords (planar graph) |
| Recursion stack | $O(k) = O(n)$ | $k \leq 2n - 4$ faces (Euler's formula) |
| Current triangulation | $O(n)$ | Chords of partial triangulation |
| **Total space** | $\mathbf{O(n)}$ | **Theorem 9** |
| **Lower Bounds (Lemma 5)** | | |
| Triangulations per face | $\Omega(n_i)$ | $d_i \geq n_i - 3$ |
| Total triangulations | $\Omega\left(\prod_{i=1}^{k} n_i\right)$ | $T = \prod_{i=1}^{k} d_i$ |

**Interpretation:**

- The algorithm spends $O(n_i)$ time building the initial root triangulation for each face $F_i$, but this cost is amortized over the $\Omega(n_i)$ triangulations generated for that face (since $d_i \geq n_i - 3$ by Lemma 5).
- Each individual triangulation is generated by a sequence of $O(1)$-time edge flip operations from its parent in the local genealogical tree.
- The bound $B + F = O(T)$ means the algorithm performs constant number of operations (chord insertions or edge flips) per output triangulation on average. Since each elementary operation takes $O(1)$ time, this gives $O(1)$ amortized time per triangulation.
- The $O(n)$ space complexity is optimal since we must store the input graph itself, which requires $\Theta(n)$ space. Importantly, space usage is independent of the number of triangulations $T$, which can be exponential in $n$.
- For graphs with $k$ faces where each face has $\Omega(n/k)$ vertices, the total number of triangulations grows exponentially: $T = \prod_{i=1}^{k} d_i \geq \prod_{i=1}^{k} (n_i - 3) = \exp(\Omega(k \log(n/k)))$. The building cost $B = O(T)$ shows perfect amortization—preprocessing work grows linearly with the exponential output size.

*Remark* 5 (Comparison with Related Work). Our result improves upon the reverse search approach [1], which typically requires $O(n^2)$ time per solution for triangulation enumeration problems. While the Parvez-Rahman-Nakano algorithm [11] achieves $O(1)$ amortized time for triconnected plane graphs, our extension to biconnected plane graphs—a strictly more general class—maintains the same optimal complexity through the novel techniques of safe root selection, local genealogical trees, and dynamic global chord tracking.

*Remark* 6 (Optimality of $O(1)$ Amortized Time). We conjecture that $O(1)$ amortized time per triangulation is optimal for this problem. The lower bound argument proceeds as follows: Since each triangulation requires $\Omega(n)$ bits to represent (specifying which of the potential $O(n^2)$ chords are present), and we must output $T$ triangulations, any algorithm requires $\Omega(nT)$ total time. This gives an $\Omega(n)$ lower bound per triangulation in the worst case.

However, amortized complexity allows preprocessing and sharing work across outputs. Our algorithm achieves $O(1)$ amortized time by:

- Generating triangulations in a tree structure where each child differs from its parent by a single $O(1)$ edge flip
- Amortizing the $O(n)$ cost of building the initial root triangulation across the $\Omega(n)$ descendants
- Using $O(1)$ hash table operations for chord conflict detection

Achieving $o(1)$ amortized time appears impossible without exploiting special structure beyond biconnectivity. For instance, sublinear amortized time would require generating multiple triangulations with a single operation, which contradicts the tree structure of the solution space where each triangulation has a unique parent (except the root).

*Remark* 7 (Practical Considerations). While our theoretical analysis uses hash sets with expected $O(1)$ operations, the practical performance can vary based on implementation details:

- **Hash function choice:** For edge/chord hashing, a simple function like $h((u, v)) = (p_1 \cdot \min(u, v) + p_2 \cdot \max(u, v)) \bmod M$ with appropriate primes $p_1, p_2$ and table size $M$ works well in practice.
- **Load factor:** Maintaining a load factor below 0.75 ensures good expected performance with reasonable space overhead.
- **Collision resolution:** Chaining with linked lists or open addressing with linear probing both perform well for the typical workload of our algorithm.

Our experimental implementation (Section 6) uses Java's built-in `HashSet` with default settings, achieving practical performance that closely matches the theoretical predictions.

# 6 Experimental Validation

## 6.1 Experimental Setup

We implemented the algorithm in C++ and conducted comprehensive benchmarking on 48 diverse biconnected plane graphs. Execution time was measured using `std::chrono::steady_clock` with nanosecond precision. Memory usage was captured via platform-specific RSS queries (`/proc/self/smaps_rollup` on Linux, `GetProcessMemoryInfo` on Windows, `task_info` on macOS). All measurements represent averages over 20 independent runs.

**System Configuration:** AMD Ryzen 5 5600G (2 cores, 3.9 GHz, 8GB RAM), Linux Mint 22.2, GCC 13.3.0 with optimization flags. Implementation uses C++ STL containers including `unordered_set` for $O(1)$ expected-time chord lookup operations.

**Test Cases:** The 48 test instances include simple cycles ($C_3$–$C_9$), wheels ($W_4$, $W_5$), complete graphs ($K_3$, $K_4$), bipartite graphs ($K_{2,3}$), diamond structures, quadrilaterals, pre-triangulated graphs, and complex general plane graphs with up to 16 vertices and over 1.2 million triangulations.

## 6.2 Performance Results

Tables 4 and 5 present detailed performance metrics. Key observations:

**Time Complexity Validation:** Time per triangulation ranges from 787ns ($K_3$) to 4,948ns ($C_4$), averaging approximately 2,100ns across all cases. The largest instance (1,282,136 triangulations) completed in 2.06 seconds at 1,606ns per triangulation, confirming $O(1)$ amortized complexity.

**Space Complexity Validation:** Memory per vertex ranges from 315B to 2KB, independent of output size. The complex 13-vertex graph generating 1.28 million triangulations uses only 315B/vertex, while small 3–4 vertex graphs show higher per-vertex overhead (512B–2KB) due to fixed data structure costs. This validates the $O(n)$ space bound with memory usage determined by input size, not exponential output size.

Table 4: Representative Performance Results (Table 4: Time & Space Analysis)

| Test Case | V | Triangulations | Time (s) | ns/Triang | Mem/Vertex |
|---|---|---|---|---|---|
| *Simple Cycles* | | | | | |
| C4 (square) | 4 | 2 | 0.000010 | 4,948 | 1.00 KB |
| C6 (hexagon) | 6 | 68 | 0.000143 | 2,109 | 682 B |
| C8 (octagon) | 8 | 4,872 | 0.007443 | 1,528 | 512 B |
| C9 (nonagon) | 9 | 46,782 | 0.066701 | 1,426 | 455 B |
| *Large-Scale* | | | | | |
| complex | 13 | 1,282,136 | 2.059353 | **1,606** | 315 B |
| graph V12E25F12 | 12 | 62 | 0.000089 | 1,437 | 682 B |
| quadrilateral 14quad | 16 | 16,384 | 0.037116 | **2,265** | 512 B |
| *Complete Graphs* | | | | | |
| K3 (triangle) | 3 | 1 | 0.000001 | 787 | 1.33 KB |
| K4 (tetrahedron) | 4 | 1 | 0.000001 | 1,369 | 1.00 KB |
| *Wheels & Bipartite* | | | | | |
| W5 (wheel) | 6 | 5 | 0.000008 | 1,662 | 682 B |
| K2_3 (bipartite) | 5 | 4 | 0.000011 | 2,627 | 819 B |

Table 5: Additional Test Cases (Table 5)

| Test Case | V | Triangulations | Time (s) | ns/Triang | Mem/Vertex |
|---|---|---|---|---|---|
| graph V9E10F3 | 9 | 13,880 | 0.025192 | 1,815 | 455 B |
| graph V8E10F4 | 8 | 40 | 0.000084 | 2,094 | 1.00 KB |
| quadrilateral 7quad | 9 | 72 | 0.000191 | 2,646 | 910 B |
| graph V7E11F5 | 7 | 7 | 0.000015 | 2,120 | 585 B |
| pentagon | 5 | 10 | 0.000031 | 3,111 | 819 B |
| diamond structures | 4 | 1 | 0.000002 | ∼1,900 | 1.00 KB |

*Full results for all 48 cases available in supplementary materials.*

## 6.3 Summary

These experimental results confirm both theoretical complexity bounds in practice:

- $O(1)$ **amortized time:** Consistent per-triangulation performance (787–4,948ns) across diverse structures, scaling to 1.28M triangulations
- $O(n)$ **space:** Memory usage (315B–2KB per vertex) independent of exponential output size, confirming linear scaling with input vertices

The implementation efficiently handles graphs ranging from simple 3-vertex triangles to complex 16-vertex structures generating millions of triangulations, validating the practical efficiency of our approach.

# 7 Conclusion and Future Work

We have presented the first algorithm for generating all triangulations of biconnected plane graphs with constant amortized time complexity. Our algorithm extends the elegant tree-of-triangulations framework of Parvez, Rahman, and Nakano [11] from triconnected to biconnected graphs by introducing safe root selection and local genealogical trees.

## 7.1 Key Achievements

1. **Theoretical Foundation:** We have provided rigorous proofs establishing:
   - Existence of at least two safe root vertices in every face (Theorem 3)
   - Completeness of the algorithm without duplications (Theorem 2)
   - Amortized $O(1)$ time complexity per triangulation (Theorem 7)
   - Linear $O(n)$ space complexity (Theorem 9)
2. **Practical Algorithm:** Efficient implementation suitable for real-world applications with extensive experimental validation
3. **Novel Techniques:** Safe root selection and global chord management strategies that can extend to other enumeration problems involving plane graph decompositions
4. **Broader Applicability:** Extension from triconnected to the more general class of biconnected plane graphs, significantly expanding the scope of graphs that can be processed efficiently

## 7.2 Theoretical Significance

This work makes several theoretical contributions:

- **Generalization:** We show that constant-time triangulation enumeration is achievable beyond the restrictive triconnected case
- **Multi-edge Resolution:** Our safe root concept provides a general strategy for handling chord conflicts in face-decomposed plane graphs
- **Amortized Analysis:** The detailed amortized analysis demonstrates that building time is dominated by output size even with complex face interactions

Each of these problems involves selecting "pivot" vertices or edges to avoid conflicts during recursive decomposition, suggesting that safe root selection strategies may have broad applicability in computational geometry.

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